



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 4 Implementation of Basic Parallel Circuit

SUBMITTED TO:

Mr. Mehmood-ul-Haq

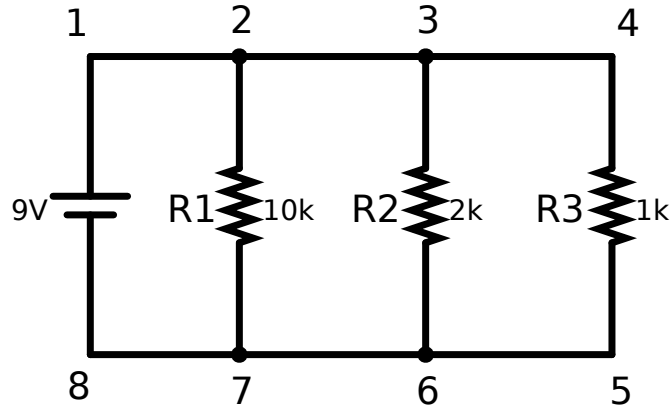
SUBMITTED BY:

Muhammad Saad (2025-CE-113)

Submission Date: 31 Mar, 2026

Department of Computer Engineering
University of Engineering and Technology Lahore
Main Campus

Experiment 1



Practical Procedure

The objective of this experiment is to analyze the behavior of a DC parallel circuit consisting of three resistors: $R_1 = 10\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, and $R_3 = 1\text{ k}\Omega$, driven by a constant **9V DC** source. For measurements, we utilize the **UNI-T UT89XE** digital multimeter to capture the individual branch currents and the common node voltage. Due to the virtual nature of the current lab sessions, the circuit is constructed and analyzed using the **Falstad CircuitJS** environment to simulate the parallel branching and current distribution.

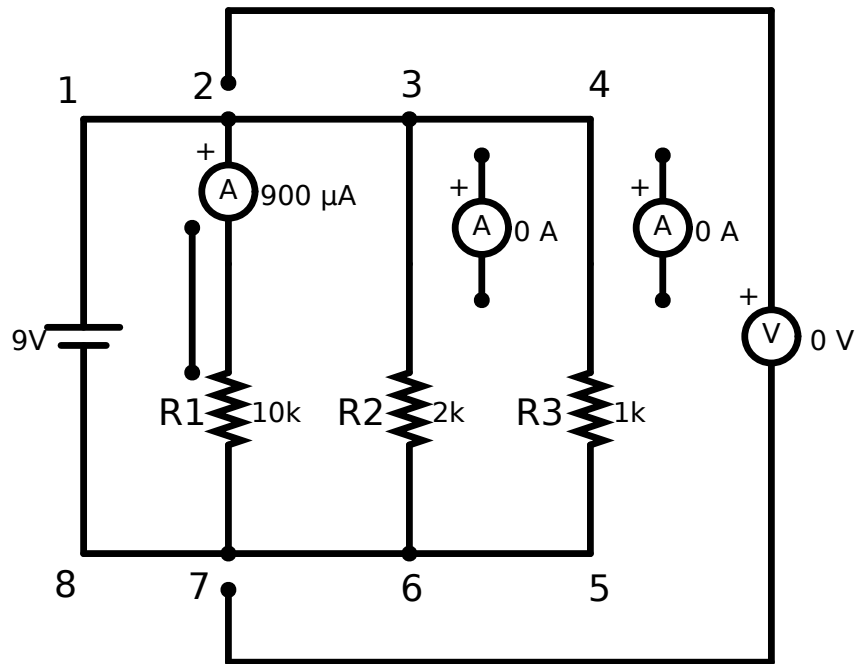
To measure the voltage between **Node 2** and **Node 7** (which represents the potential difference across the parallel network), the multimeter dial is set to the **20V DC** range (DC indicated by the **V—** symbol). The black probe is inserted into the **COM** port while the red probe is placed in the **VΩHz** terminal. The probes are placed in **parallel** across the nodes. For the branch current measurements (I_{R1} , I_{R2} , and I_{R3}), the red probe must be relocated to the **mA μA** terminal. Depending on the expected magnitude, the dial is set to the **2000μA** or **200mA DC** range (DC indicated by the **A—** symbol). Measuring branch current requires the DMM to be connected in **series** with each individual resistor by breaking the specific branch and inserting the meter into the path.

Theoretical calculations for parallel circuits assume that the voltage remains perfectly constant across all branches; however, simulated and practical results may exhibit slight variations. This is primarily influenced by the **internal resistance** of the ammeter, which introduces a small **burden voltage** when inserted into a branch, potentially shifting the current distribution. Additionally, the **tolerance** of the physical resistors and the precision of the **9V** supply calibration contribute

to the minor discrepancies observed between the mathematical predictions and the simulated data recorded in this lab.



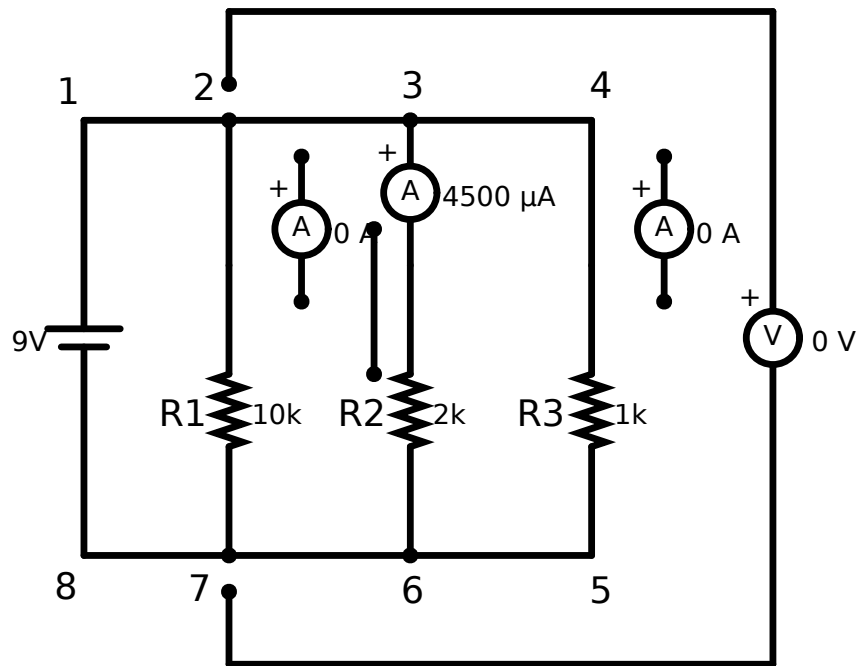
DMM: Dial to
2000 μ A DC



Falstad: Current through R_1



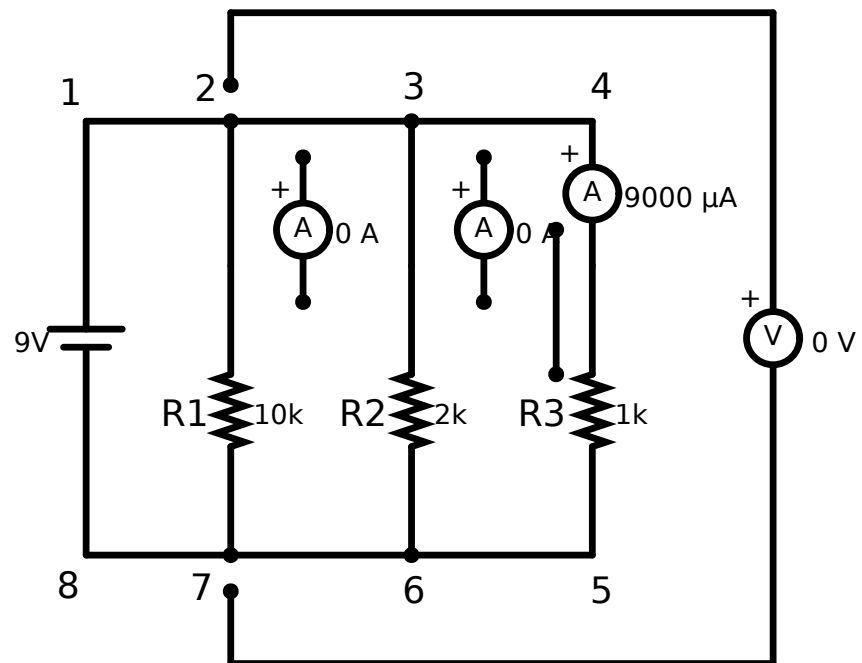
DMM: Dial to
200mA DC



Falstad: Current through R_2



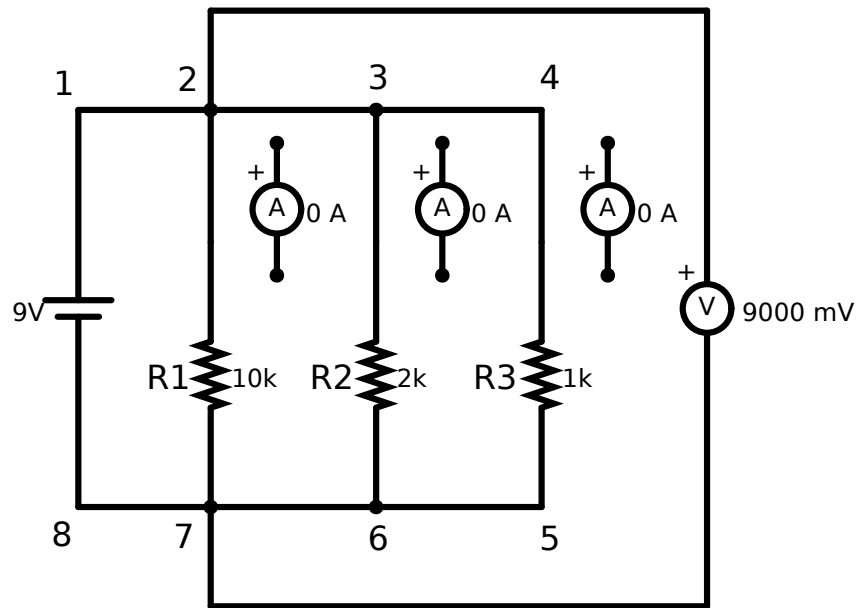
DMM: Dial to
200mA DC



Falstad: Current through R_3



DMM: Dial
to **20V DC**



Falstad: Voltage between Node 2 and Node 7

Theoretical Calculations

The following calculations assume an ideal DC source with $V_{in} = 9\text{ V}$. In a parallel circuit, the voltage across each branch is equal to the source voltage: $V_{in} = V_{R1} = V_{R2} = V_{R3} = 9\text{ V}$.

Total Resistance, R_{total}

For the parallel circuit containing $R_1 = 10\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, and $R_3 = 1\text{ k}\Omega$:

$$\frac{1}{R_{total}} = \frac{1}{10\text{ k}\Omega} + \frac{1}{2\text{ k}\Omega} + \frac{1}{1\text{ k}\Omega} = \frac{1}{1000} \cdot \left(\frac{1}{10\text{ }\Omega} + \frac{1}{2\text{ }\Omega} + \frac{1}{1\text{ }\Omega} \right)$$

$$\frac{1}{R_{total}} = \frac{1}{1000} \cdot (0.1 + 0.5 + 1.0) = 1.6\text{ mS} \Rightarrow R_{total} = \frac{1}{1.6\text{ mS}} = 0.625\text{ k}\Omega$$

Current through R_1 , I_{R1}

$$I_{R1} = \frac{V_{in}}{R_1} = \frac{9\text{ V}}{10\text{ k}\Omega} = 0.9\text{ mA}$$

Current through R_2 , I_{R2}

$$I_{R2} = \frac{V_{in}}{R_2} = \frac{9\text{ V}}{2\text{ k}\Omega} = 4.5\text{ mA}$$

Current through R_3 , I_{R3}

$$I_{R3} = \frac{V_{in}}{R_3} = \frac{9\text{ V}}{1\text{ k}\Omega} = 9.0\text{ mA}$$

Total Circuit Current, I_{total}

According to Kirchhoff's Current Law (KCL), the total current is the sum of individual branch currents:

$$I_{total} = I_{R1} + I_{R2} + I_{R3} = 0.9\text{ mA} + 4.5\text{ mA} + 9.0\text{ mA} = 14.4\text{ mA}$$

Voltage between Node 2 and Node 7

Since Node 2 and Node 7 are connected directly across the parallel network and the power supply:

$$V_{2-7} = V_{in} = 9.0\text{ V}$$

Experiment 1 Data

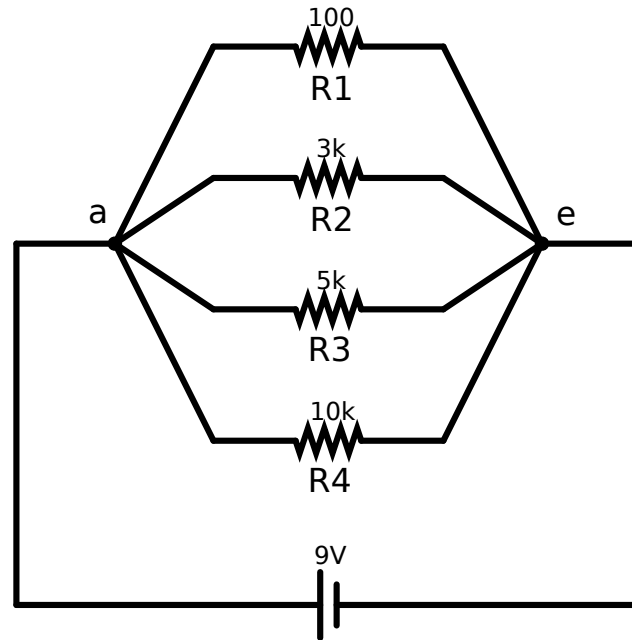
Practical Values (Simulated)

R_1 Current	R_2 Current	R_3 Current	Voltage between nodes 2 and 7
0.900 mA	4.500 mA	9.000 mA	9.000 V

Theoretical Values

R_1 Current	R_2 Current	R_3 Current	Voltage between nodes 2 and 7
0.900 mA	4.500 mA	9.000 mA	9.000 V

Experiment 2



Practical Procedure

The objective of this experiment is to analyze the behavior of a DC parallel circuit consisting of four resistors: $R_1 = 100\ \Omega$, $R_2 = 3\ \text{k}\Omega$, $R_3 = 5\ \text{k}\Omega$, and $R_4 = 10\ \text{k}\Omega$, driven by a constant **9V DC** source. For measurements, we utilize the **UNI-T UT89XE** digital multimeter to capture the individual branch currents and the potential difference between the main supply nodes. Due to the virtual nature of the current lab sessions, the circuit is constructed and analyzed using the **Falstad CircuitJS** environment to simulate the parallel branching and current distribution.

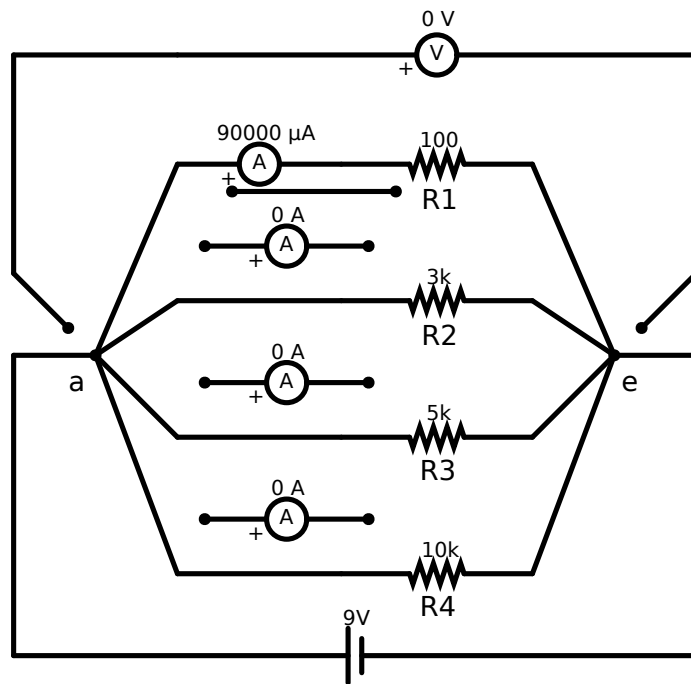
To measure the voltage between **Node a** and **Node e** (which represents the potential difference across the parallel network), the multimeter dial is set to the **20V DC** range (DC indicated by the **V—** symbol). The black probe is inserted into the **COM** port while the red probe is placed in the **VΩHz** terminal. The probes are placed in **parallel** across the nodes. For the branch current measurements (I_1 through I_4), the red probe must be relocated to the **mA μA** terminal. Depending on the expected magnitude, ranging from high current at the $100\ \Omega$ branch to low current at the $10\ \text{k}\Omega$ branch, the dial is adjusted between the **2000μA** and

200mA DC ranges (DC indicated by the **A**— symbol). Measuring branch current requires the DMM to be connected in **series** with each individual resistor by breaking the specific branch and inserting the meter into the path.

Theoretical calculations for parallel circuits assume that the voltage remains perfectly constant across all branches; however, simulated and practical results may exhibit slight variations. This is primarily influenced by the **internal resistance** of the ammeter, which introduces a small **burden voltage** when inserted into a branch, potentially shifting the current distribution. Additionally, the **tolerance** of the physical resistors and the precision of the **9V** supply calibration contribute to the minor discrepancies observed between the mathematical predictions and the simulated data recorded in this lab.



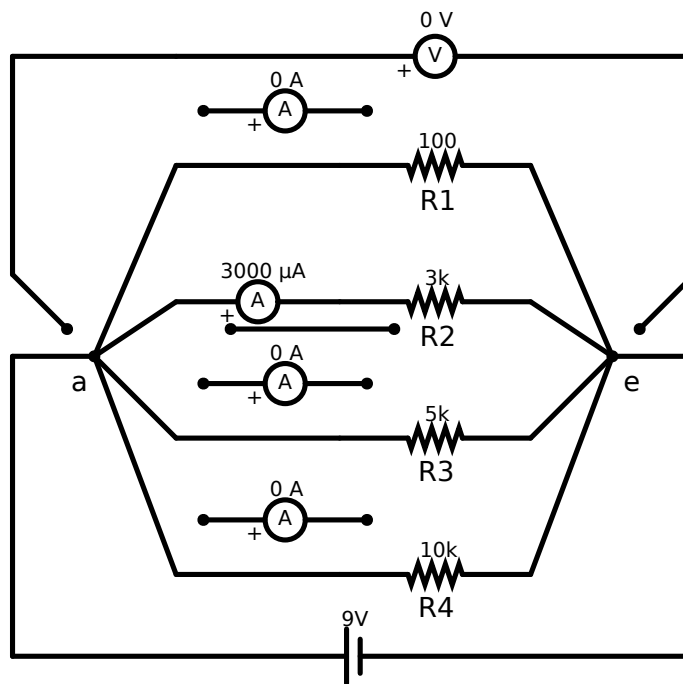
DMM: Dial to
200mA DC



Falstad: Current through R_1



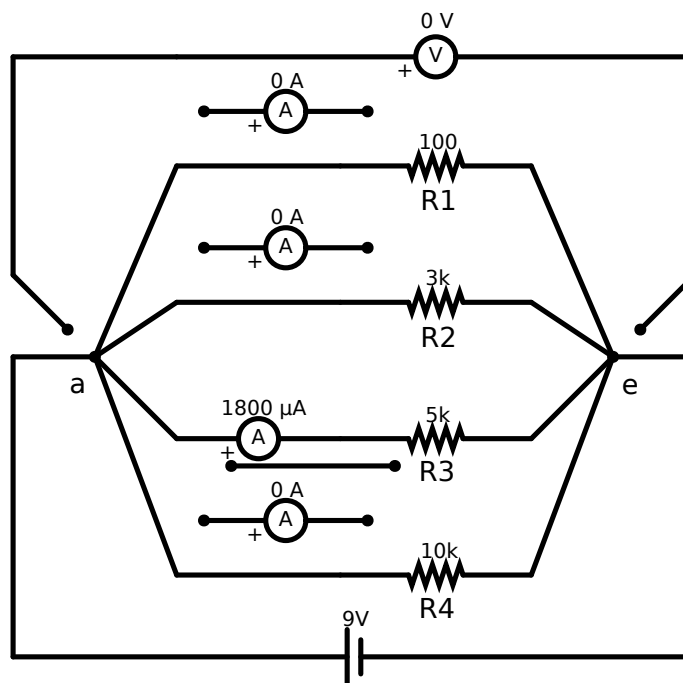
DMM: Dial to
200mA DC



Falstad: Current through R_2



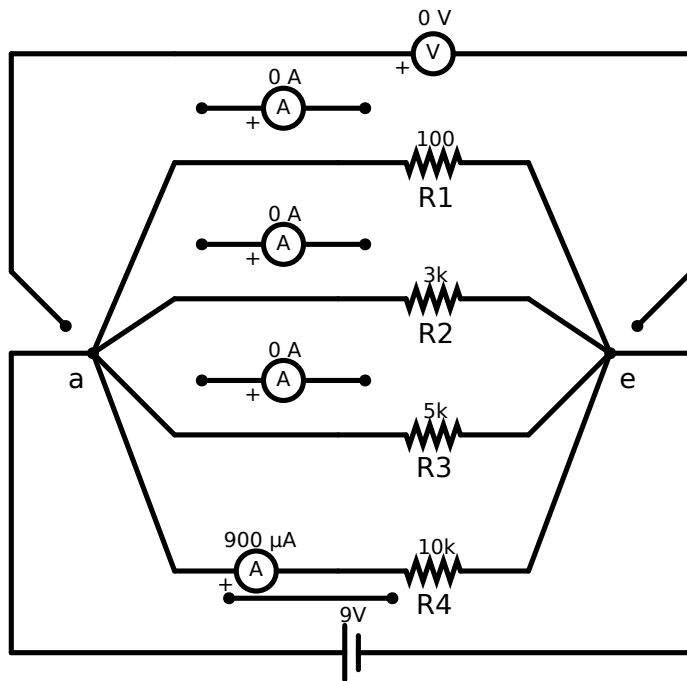
DMM: Dial to
2000μA DC



Falstad: Current through R_3



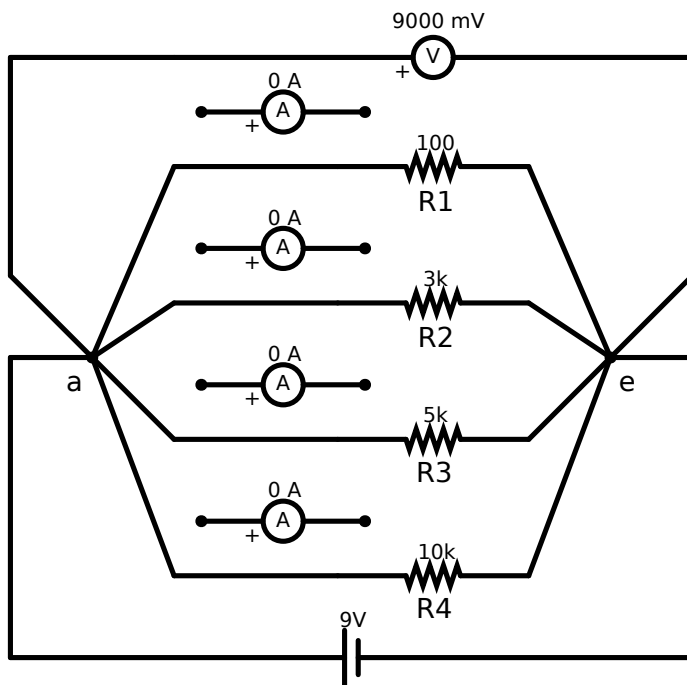
DMM: Dial to
2000µA DC



Falstad: Current through R_4



DMM: Dial
20V DC



Falstad: Voltage between Node a and Node e

Theoretical Calculations

The following calculations assume an ideal DC source with $V_{in} = 9\text{ V}$. In this parallel network, the voltage across every branch remains constant and equal to the source: $V_{in} = V_{R1} = V_{R2} = V_{R3} = V_{R4} = 9\text{ V}$.

Total Resistance, R_{total}

For the parallel circuit containing $R_1 = 100\ \Omega$, $R_2 = 3\text{ k}\Omega$, $R_3 = 5\text{ k}\Omega$, and $R_4 = 10\text{ k}\Omega$:

$$\begin{aligned}\frac{1}{R_{total}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \\ \frac{1}{R_{total}} &= \frac{1}{100\ \Omega} + \frac{1}{1000} \cdot \left(\frac{1}{3\ \Omega} + \frac{1}{5\ \Omega} + \frac{1}{10\ \Omega} \right) \\ \frac{1}{R_{total}} &= 0.01 + \frac{1}{1000} \cdot (0.33333 + 0.2 + 0.1) = 0.01 + 0.00063 = 0.01063\text{ S}\end{aligned}$$

$$R_{total} = \frac{1}{0.01063\text{ S}} \approx 94.04\ \Omega$$

Current through R_1 , I_1

$$I_1 = \frac{V_{in}}{R_1} = \frac{9\text{ V}}{100\ \Omega} = 90.0\text{ mA}$$

Current through R_2 , I_2

$$I_2 = \frac{V_{in}}{R_2} = \frac{9\text{ V}}{3\text{ k}\Omega} = 3.0\text{ mA}$$

Current through R_3 , I_3

$$I_3 = \frac{V_{in}}{R_3} = \frac{9\text{ V}}{5\text{ k}\Omega} = 1.8\text{ mA}$$

Current through R_4 , I_4

$$I_4 = \frac{V_{in}}{R_4} = \frac{9\text{ V}}{10\text{ k}\Omega} = 0.9\text{ mA}$$

Total Circuit Current, I_{total}

Using Kirchhoff's Current Law (KCL), the total current entering the parallel network is:

$$I_{total} = I_1 + I_2 + I_3 + I_4 = 90.0 + 3.0 + 1.8 + 0.9 = 95.7 \text{ mA}$$

Voltage between Node a and Node e

As Node a and Node e are the primary supply nodes for the parallel branches:

$$V_{a-e} = V_{in} = 9.0 \text{ V}$$

Experiment 2 Data

Practical Values (Simulated)

R_1 Current	R_2 Current	R_3 Current	R_4 Current	Voltage between nodes a and e)
90.000 mA	3.000 mA	1.800 mA	0.900 mA	9.000 V

Theoretical Values

R_1 Current	R_2 Current	R_3 Current	R_4 Current	Voltage between nodes a and e
90.000 mA	3.000 mA	1.800 mA	0.900 mA	9.000 V

Post-Lab Questions

1. Why is the value of voltage across each resistor in a parallel circuit the same?

In a parallel circuit, all components are connected directly between the same two common nodes. Because voltage (potential difference) is defined as the difference in electrical potential between two points, and every resistor in a parallel network shares the exact same start and end points, they must experience the same electrical pressure. Mathematically, there is no impedance between the nodes along the connecting wires in an ideal model, ensuring $V_{source} = V_1 = V_2 = V_n$.

2. When parallel resistors are of three different values, which has the greatest power loss?

The resistor with the **lowest ohmic value** will experience the greatest power loss. Electrical power can be expressed by the formula $P = V^2/R$. Since the voltage (V) is constant for all branches in parallel, power is inversely proportional to resistance. Therefore, the branch with the least resistance draws the most current and, consequently, dissipates the most energy as heat.

3. How does the current across any branch of a parallel circuit vary?

The current in a parallel branch varies **inversely** with the resistance of that specific branch, according to Ohm's Law ($I = V/R$). While the total voltage remains fixed, the charge carriers will preferentially follow the path of least resistance. If the resistance of a branch increases, the current through it decreases; if the resistance decreases, the current increases.

4. What happens to total resistance in a circuit with parallel resistors if one of them is opened (i.e., removed)?

If a resistor is removed or "opened," the total equivalent resistance of the circuit **increases**. In a parallel configuration, each additional resistor provides an extra "path" for current, which reduces the overall opposition to flow. When a branch is removed, one of these paths is eliminated, forcing the total current to decrease for the same source voltage, which mathematically results in a higher R_{total} .